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Suspension compliance to reduce frame loading

Abstract

A harvesting system includes a header pivotally attached to a combine. The header includes a center section to which a left wing and right wing are pivotally attached. A suspension system of the harvesting system includes first and second engageable states that enable dynamic wing behavior and reduce structural load. The first state corresponds to a harvesting configuration of the header in which the wings are allowed to pivot to allow the header to follow changes in terrain. The second state corresponds to a configuration in which the header is elevated relative to the ground. In the second state, the ability of the wings to pivot is minimized as compared to the first state, which allows the header to be maintained in a substantially flat configuration while minimizing the amount of dynamic load imparted by the header on the combine during non-harvesting transport of the header.

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Background/Summary

CROSS-REFERENCE TO RELATED PATENT APPLICATIONS (1) This application is a divisional of U.S. application Ser. No. 17/686,189, filed Mar. 3, 2022, which is a continuation of U.S. application Ser. No. 16/414,573, filed May 16, 2019, which claims the benefit and priority under 35 U.S.C. § 119(e) to U.S. Provisional Application No. 62/693,278, titled “SUSPENSION COMPLIANCE TO REDUCE FRAME LOADING,” filed Jul. 2, 2018, each of which are incorporated herein by reference in their entirety.

BACKGROUND

(1) Crop harvesting is commonly performed by a harvesting system comprising a combine harvester (“combine”) equipped with a removable header designed for harvesting crops. In an attempt to increase the throughput of such harvesting systems, combines are being paired with increasingly wider headers. However, although the increased span of such wider headers may improve throughput by increasing the rate at which ground can be covered by the harvesting system,

the increased width of the header may result in a decrease in crop yield efficiency. In particular, given the rigid, flat configuration of headers typically used in such harvesting systems, the increased inability of wider, rigid frame header to conform to variations in terrain often results in a decrease in the amount of crop that is harvested as the harvesting system travels over uneven terrain.

(2) Additionally, increasing the width of the header of a harvesting system often increases the structural loads imparted by the heavier, wider header onto the combine. As a result, many combines that are used in such wider header harvesting systems incorporate reinforced combine structures configured to support the added weight of a wider header and to withstand and resist the increased dynamic loads that such wider headers impart. In addition to increasing the material costs required to manufacture such reinforced combines, the added mass of such reinforced combines also typically increases the costs of operating the harvesting system.

(3) One implementation of the present disclosure is a harvesting system that includes a harvesting combine and a harvesting header. The harvesting header includes a center section, a left wing hingedly attached to the center section, and a right wing hingedly attached to the center section. The header is pivotally attached to a support structure of the harvesting combine. The header is configured to pivot relative to the support structure to transition between a first position in which the header is at least partially supported by a ground surface and a second position in which the header is entirely supported by the combine. When the header is in the first position, the ability of the left wing and the right wing to pivot upward or downward relative to the center section is constrained to a first operating range. When the header is in the second position, the ability of the left wing and the right wing to pivot upward or downward relative to the center section is constrained to a second operating range. The first operating range is greater than the second operating range.

(4) According to some embodiments, the second operating range is greater than 0° . According to some embodiments, the first operating range corresponds to a range that extends between approximately 5.0° upwards and approximately 5.0° downwards as measured relative to the center section. According to some embodiments, the second operating range is approximately 20% of the first operating range. According to some embodiments, the second operating range corresponds to an angular displacement of approximately 2.0° .

(5) According to some embodiments, the harvesting system further includes a variable spring suspension system configured to selectively constrain the pivoting of the left wing and right wing relative to the center section to either the first operating range or the second operating range.

(6) One implementation of the disclosure is a harvesting system including a harvesting combine, a harvesting header and a suspension system. The harvesting header includes a center section, a left wing hingedly attached to the center section, and a right wing hingedly attached to the center section. The suspension system has an engageable first variable state and an engageable second variable state. The second variable state is different than the first variable state. When the header is in a harvesting position and the first variable state of the suspension system is engaged, the left wing and right wing are allowed to pivot relative to the center section. When the header is elevated entirely off of the ground, the second variable state of the suspension system is automatically engaged and the left wing and the right wing are allowed to pivot upward or downward relative to the center section. When the header is elevated entirely off of the ground, the second variable state of the suspension system is automatically engaged and the left wing and the right wing are allowed to pivot upward or downward relative to the center section.

(7) According to some embodiments, the degree to which the left wing and right wing are allowed to pivot relative to the center section when the header is in the harvesting position is greater than the degree to which the left wing and right wing are allowed to pivot when the header is elevated entirely off of the ground.

(8) According to some embodiments, when the first variable state of the suspension system is

engaged, the right wing and left wing are allowed to pivot within a range of approximately 5.0° upwards and approximately 5.0° downwards as measured relative to the center section. According to some embodiments, when the second variable state of the suspension system is engaged, the right wing and left wing are allowed to pivot within a range that corresponds to an angular displacement of approximately 2.0° . According to some embodiments, when the second variable state of the suspension system is engaged, the left wing and right wing are allowed to pivot within a range that is approximately 20% less than the range within which the left wing and right wing are allowed to pivot when the first variable state of the suspension system is engaged.

(9) According to some embodiments, directly subsequent to the second variable state of the suspension system being engaged, the position of the left wing relative to the center section corresponds to the position of the left wing relative to the center section directly prior to the second variable state of the suspension system being engaged, and the position of the right wing relative to the center section corresponds to the position of the right wing relative to the center section directly prior to the second variable state of the suspension system being engaged.

(10) According to some embodiments, the suspension system includes a fluid cylinder fluidly connected to an accumulator and a valve selectively openable to permit flow between the fluid cylinder and the accumulator. The suspension system is configured to engage the second variable state by closing the valve to block flow between the fluid cylinder and accumulator. According to some embodiments, the fluid cylinder is fluidly connected to the accumulator by an attenuation hose. According to some embodiments, fluid from the fluid cylinder is configured to flow into the attenuation hose when the second variable state of the suspension system is engaged.

(11) One implementation of the present disclosure is a harvester system including a harvesting combine, and articulated header, and a variable spring suspension system. The articulated header is defined by at least a first section and a second section. The variable spring suspension system has a first variable spring state and a second variable spring state. In the first variable spring state, the suspension system is configured to allow the first section and second section to pivot relative to one another according to a first range of motion. In the second variable spring state, the suspension system is configured to allow the first section and second section to pivot relative to one another according to a second range of motion. The first range of motion is greater than the second range of motion.

(12) According to some embodiments, the header is pivotally attached to the combine and configured to move between a first position in which at least a portion of the header weight is supported by the ground and a second position in which an entirety of the header weight is supported by the combine. The suspension system is configured to activate the first variable spring state when the header is in the first position and to activate the second variable spring state when the header is in the second position.

(13) According to some embodiments, the harvester system further includes a control system having an automatic header height control mode. The suspension system is configured to automatically activate the second variable spring state in response to the header height control mode being deactivated.

(14) According to some embodiments, the suspension system includes a fluid cylinder fluidly connected to an accumulator and a valve selectively openable to permit flow between the fluid cylinder and the accumulator. The suspension system is configured to activate the second variable spring state by closing the valve to block flow between the fluid cylinder and accumulator.

(15) According to some embodiments, the suspension system includes a first coiled spring and a second coiled spring. The first coiled spring is configured to impart a greater spring force than the second coiled spring. The suspension system is configured to selectively engage the first coiled spring to impart a spring force on the first section when the header is in the second position.

(16) Those skilled in the art will appreciate that the summary is illustrative only and is not intended to be in any way limiting. Other aspects, inventive features, and advantages of the devices and/or

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIGS. **1A-1C** illustrate a harvester in a harvesting configuration travelling over different types of terrain, according to some embodiments.
- (2) FIG. **2** is a simplified block diagram illustrating a top view of a header, according to some embodiments.
- (3) FIG. **3** illustrates a harvester in a non-harvesting transport configuration, according to some embodiments.
- (4) FIG. **4** is a simplified diagram illustrating a suspension system, according to some embodiments.
- (5) FIGS. **5A-5C** are simplified general block diagrams illustrating a blocker valve, according to some embodiments.
- (6) FIG. **6A** is a simplified general block diagram illustrating a suspension system during a harvesting configuration of a harvester, according to some embodiments.
- (7) FIG. **6B** is a simplified general block diagram illustrating a front view of a header during the harvesting configuration of the harvester shown in FIG. **6A**, according to some embodiments.
- (8) FIG. **7A** is a simplified general block diagram illustrating a suspension system during a transition configuration of a harvester, according to some embodiments.
- (9) FIG. **7B** is a simplified general block diagram illustrating a front view of a header during the transition configuration of the harvester shown in FIG. **7A**, according to some embodiments.
- (10) FIG. **8A** is a simplified general block diagram illustrating a suspension system during a non-harvesting transport configuration of a harvester, according to some embodiments.
- (11) FIG. **8B** is a simplified general block diagram illustrating a front view of a header during the non-harvesting transport configuration of the harvester shown in FIG. **8A**, according to some embodiments.
- (12) FIG. **9A** is a simplified general block diagram illustrating a suspension system during a downward flex configuration of a harvester, according to some embodiments.
- (13) FIG. **9B** is a simplified general block diagram illustrating a front view of a header during the downward flex configuration of the harvester shown in FIG. **9A**, according to some embodiments.

DETAILED DESCRIPTION

(14) Referring generally to the FIGURES, a suspension system **200** for a harvester **100** configured to reduce structural loads is shown. As will be described in more detail below, suspension system **200** is configured as a variable spring rate suspension system, which allows the header **104** to more closely and easily follow terrain while the harvester **100** is in a harvesting mode, while also providing the header **104** with the ability to flex during an elevated, non-harvesting transport configuration of the header **104**. In doing so, the suspension system **200** reduces the structural loads that the combine **102** supporting the header **104** is subject to during operation of the harvester **100**. As such, the suspension system **200** allows the width of the header **104** to be increased (so as to, e.g., increase harvesting throughput) without requiring reinforcement of the structure of the combine **102** to support the increased mass of the wider header **104**.

(15) Referring to FIGS. **1A-1C**, an agricultural harvester **100** according to one embodiment is shown in various harvesting configurations as the harvester **100** travels over terrain having varying contours. As illustrated in FIGS. **1A-1C**, according to various embodiments, the harvester **100** includes a combine **102** and an agricultural harvesting header **104** supported on the front of the combine **102**.

(16) As illustrated by the simplified block diagram of FIG. 2, according to various embodiments, the header **104** defines an articulated structure comprising a center section **142** to which a left wing **144a** is hingedly connected by a left hinge joint **146a** and to which a right wing **144b** is hingedly connected by a right hinge joint **146b**. The connection of the left wing **144a** to the center section **142** via left hinge joint **146a** allows the left wing **144a** to pivot upwards or downwards relative to the center section **142** about a generally horizontal axis along which the left hinge joint **146a** extends. Similarly, the connection of the right wing **144b** to the center section **142** via right hinge joint **146b** allows the right wing **144b** to pivot upwards or downwards relative to the center section **142** about a generally horizontal axis along which the right hinge joint **146b** extends. As will be understood, given the independent hinged attachment of each of the left wing **144a** and the right wing **144b** to the center section **142**, the left wing **144a** may pivot in any direction (i.e. upwardly or downwardly) and to any degree, irrespective of any pivoting of the right wing **144b** about the center section **142**, and vice versa.

(17) Although, as described below, the harvester **100** comprises a suspension system **200** configured to maintain the header **104** in a generally flat configuration, according to some embodiments, such as, e.g., illustrated in FIG. 2, a manually or automatically actuated lock **148** may be provided between left wing **144a** and center section **142** and/or between right wing **144b** and center section **142** which may optionally be used in situations in which a user may desire to fixedly and rigidly restrain the pivoting movement of left wing **144a** and/or right wing **144b** relative to the center section **142**.

(18) As the harvester **100** transitions from travelling along generally flat terrain (during which the center section **142**, left wing **144a** and right wing **144b** each extend along a generally horizontal plane, such as, e.g., illustrated in FIG. 1A, that is substantially parallel to the terrain on which the header **104** is supported) to uneven terrain, the hinged connections of the left wing **144a** and right wing **144b** to the center section **142** allow the header **104** to more closely adapt to and conform to the contours of the variable terrain (such as, e.g., illustrated in FIGS. 1B and 1C).

(19) In addition to increasing crop yield, by allowing the left wing **144a** and right wing **144b** to independently flex and adapt to changing terrain, the mass of the header **104** that is accelerated as the header **104** travels over uneven terrain is decreased, thereby minimizing the structural loads on the combine **102**. Thus, the articulated configuration of the header **104** allows the width of the header **104** to be increased (as compared to a rigid, non-articulated header) without necessarily resulting in increased stress on the combine **102**, thereby obviating the need to reinforce the combine **102** to support the wider width header **104**.

(20) The combine **102** generally includes a combine harvester vehicle **106** and feederhouse **108** pivotally attached about a rear end to a lower portion of the combine harvester vehicle **106** (such as, e.g., to a chassis of the harvester vehicle **106**). A forward end of the feederhouse **108** is configured to support the header **104**. According to various embodiments, one or more feederhouse actuators (not shown) are operably coupled between the rear end of the feederhouse **108** and the combine harvester vehicle **106**. The feederhouse actuators may comprise any number of known actuator arrangements, with selective manual and/or automatic activation of the feederhouse actuator(s) being configured to cause the rear end of the feederhouse **108** to pivot relative to the combine harvester vehicle **106**, thereby resulting in a vertical movement of the forward end of the feederhouse **108**, as well as the resultant vertical movement of the attached header **104**, in an upwards or downwards direction, such as illustrated, e.g., in FIG. 3.

(21) As will be understood, the activation of these feederhouse actuators may allow the harvester **100** to transition between a harvesting configuration in which the weight of the header **104** is at least partially supported by the ground, to a non-harvesting transport configuration in which the header **104** is elevated with reference to the ground (and in which configuration the weight of the header **104** is supported entirely by the combine **102**), such as, e.g., illustrated in FIG. 3.

(22) In light of the articulated configuration of the header **104**, when feederhouse actuator(s) are

activated to raise the header **104** to an elevated, non-harvesting transport configuration such as shown in FIG. **3**, the hinged attachment of left wing **144a** and right wing **144b** (“wings **144**”) to center section **142** via left hinge joint **146a** and right hinge joint **146b**, respectively, may cause the outermost ends of wings **144** to sag relative to the height of center section **142**. As will be understood, the amount of downward displacement or sag of the outermost ends of the wings **144** as measured relative to the center section **142** increases as the width of the wings **144** is increased. (23) As described above, the ability of wings **144** to pivot substantially relative to center section **142** may advantageously allow the header **104** to conform to the terrain during harvesting.

However, such substantial pivoting movement of the wings **144** relative to the center section **142** may be undesirable when the header **104** is in an elevated position (e.g., when the harvester **100** is being turned around on end rows or during non-harvesting transport of the harvester **100**). In particular, leaving the wings **144** unsupported and free to pivot relative to center section **142** while the header **104** is elevated may cause the outermost ends of wings **144** to fully lower, thereby decreasing clearance to the ground even when the header **104** is in a fully raised configuration, which may allow inadvertent contact between the ground and header **104** that could damage the header **104**.

(24) Although preventing sagging of the outermost ends of the wings **144** in order to maintain a substantially flat profile of the header **104** may be desirable when the header **104** is in an elevated configuration such as, e.g., illustrated in FIG. **3**—for reasons as described with reference to rigid frame, non-articulated headers above—it may be undesirable to lock or otherwise fix the wings **144** into a substantially rigid configuration in an attempt to prevent the outermost ends of wings **144** from doing so. In particular, locking or otherwise restricting movement of the wings **144** relative to the center section **142** during an elevated configuration of the header **104** (such as, e.g., during non-harvesting transport of the harvester **100**) may undesirably increase the dynamic loads that are imparted by the header **104** onto the combine **102**.

(25) Instead, as will be described in more detail below, the harvester **100** is advantageously provided with a suspension system **200** that allows for some degree of pivoting movement of the wings **144** of the header **104** relative to the center section **142** while also supporting the header **104** in a substantially flat profile during field transport of the harvester **100** (i.e. when the header **104** is lifted entirely off of the ground). In doing so, the suspension system **200** minimizes the amount of header **104** inertia that must be accelerated when encountering bumps in terrain, thereby reducing the forces imparted on the combine **102** during travel of the harvester **100** with the header **104** in an elevated configuration.

(26) Accordingly, in various embodiments, the harvester **100** is provided with a variable spring rate suspension system **200** configured to be capable of being operated in a transport mode providing an increased degree of resistance to the displacement of the wings **144** relative to the center section **142** when the header **104** is in an elevated configuration (so as to, e.g., minimize or prevent the amount of downward displacement of the outermost ends of the wings **144** during non-harvesting transport of the harvester **100**) and a harvesting mode providing a decreased degree of resistance to the displacement of the wings **144** relative to the center section **142** when the header **104** is in a harvesting configuration, so as to allow the hingedly attached wings **144** to pivot as needed relative to the center section **142** while the harvester **100** is in a harvesting configuration (i.e. when the header **104** is at least partially supported along the ground), thus minimizing the structural loading of the combine **102** by the header **104**. As such, the suspension system **200** may allow the harvester **100** to incorporate a wider header **104** for more efficient harvesting throughput without requiring a reinforced combine **102** structure to support the wider width header **104**.

(27) According to various embodiments, when the harvester **100** is in a harvesting configuration (i.e. when the header **104** is at least partially supported by the ground, such as, e.g., illustrated in FIGS. **1A-1C**) and the suspension system **200** is engaged in the harvesting mode, the suspension system **200** of the harvester **100** may exhibit a degree of stiffness that is configured to allow for

upward and downward pivoting of the wings **144** by approximately no more than $\pm 20.0^\circ$, more specifically by approximately no more than $\pm 10.0^\circ$, and more specifically by approximately no more than $\pm 5.0^\circ$ as measured relative to the lateral axis along which the center section **142** extends. When the harvester **100** is in an elevated, non-harvesting transport position (i.e. when the header **104** is elevated such that the mass of the header **104** is not supported by the ground) and the suspension system **200** is engaged in the transport mode, the degree stiffness of the suspension system **200** may be greater as compared to the stiffness characterizing the suspension system **200** in the harvesting mode, such that the upwards or downwards pivoting of the wings **144** is constrained to between approximately 10% and 30%, more specifically between approximately 15% and 25%, and even more specifically approximately 20% of the range through which the wings **144** are allowed to pivot when the harvester **100** is in the harvesting configuration. According to various embodiments, the upward and downward pivoting of the wings **144** as measured relative to the lateral axis along which the center section **142** extends is approximately no more than $\pm 15.0^\circ$, more specifically is no more than approximately $\pm 7.5^\circ$, and even more specifically no more than approximately $\pm 5.0^\circ$ when the harvester **100** is in a non-harvesting transport position (such as, e.g., illustrated in FIG. 3). By defining differing suspension spring rates in each of the harvesting and non-harvesting transport modes, the suspension system **200** is configured to allow for between an approximately 10% to approximately 20% reduction in the stress imparted onto the combine **102** by the articulated header **104**, as compared to the structural load that would be imparted by a rigid, non-articulated header **104** having a similar width and mass.

(28) As will be understood, the suspension system **200** may be defined by any number of and combination of different components that are arranged in a manner to allow for the selective change in spring rate of the suspension system **200** in the different operating modes, so as to, e.g., allow for the selective constraint of the movement of the wings **144** relative to the center section **142** according to first and second variable states. In particular, in a first variable state corresponding to the harvesting mode, the suspension system **200** is defined by a first stiffness that may, according to some embodiments, be configured such that movement of the wings **144** is constrained to a first range (such as, e.g., described with reference to the harvesting configuration above). Meanwhile, in a second variable state corresponding to a transport mode, the suspension system **200** is defined by a second stiffness that may, according to some embodiments, be configured such that movement of the wings **144** is constrained to a second range that is less than the first range (such as, e.g., described with reference to the non-harvesting transport configuration above).

(29) For example, according to some embodiments (not shown), suspension system **200** may comprise a first set of coiled springs positioned about the left wing **144a** and a second set of coils positioned about right wing **144b**. Each of the first set and second set of coils comprise a first spring and a second spring. One or both of the length of the first spring and/or spring constant of the first spring differs from the second spring, such that the spring force of the first spring is greater than the spring force of the second spring. The first and second springs are configured to be independently engageable, such that, when the suspension system **200** is operated in the transport mode, the first spring is engaged, and the stiffness of the suspension system **200** is increased as compared to the stiffness of the suspensions system **200** during operation in the harvesting mode. Accordingly, in the transport mode the pivoting movement of the wings **144** about center section **142** of the header **104** may be constrained to a smaller range of motion than when the suspension system **200** is operated in the harvesting mode in which the first spring is disengaged, and the second spring is engaged.

(30) Accordingly, in such embodiments, by selectively engaging the second springs, the decreased spring rate of the suspension system **200** in the harvesting mode may provide the wings **144** with sufficient ability to pivot about center section **142** so as to allow the wings **144** to adapt to the contours of changing terrain when the harvester **100** is in harvesting position. Meanwhile, by

selectively engaging the first springs, the increased spring rate of the suspension system **200** in the non-harvesting transport mode may be configured to result in a more constrained movement of the wings **144** relative to the center section **142**, thereby minimizing the degree of displacement of the outermost ends of the wings **144** relative to the center section **142** (and thereby minimizing the risk of the outermost ends inadvertently contacting the ground when the header **104** is in an elevated, transport position) while also providing the wings **144** with sufficient flexibility to pivot so as to minimize the dynamic loads on the combine **102** during non-harvesting transport of the harvester **100** (such as, e.g., illustrated in FIG. 3).

(31) Alternatively, in other coiled spring embodiments of suspension system **200**, a single coiled spring may be positioned about each of the left wing **144a** and the right wing **144b**. The suspension system **200** may further comprise a length adjusting mechanism associated with each of the left wing **144a** and right wing **144b**, which is selectively actuatable to increase or decrease the effective length of the coiled spring. During non-harvesting transport with the header **104** in an elevated transport position, the length adjusting mechanisms may be actuated to effectively shorten the lengths of the springs, thereby increasing the spring rate of the springs (and resultant stiffness of the suspension system **200**) and thereby increasing the resistance to the movement of the wings **144** relative to center section **142**. Meanwhile, engagement of the harvesting mode of the suspension system **200** may cause the length adjusting mechanisms to be actuated to effectively lengthen the springs, thereby decreasing the spring rate of the springs (and resultant stiffness of the suspension system **200**) and thereby decreasing the degree of resistance to the movement of the wings **144** relative to the center section **142**. As will be understood, according to various embodiments, the length adjusting mechanisms may be configured to allow the effective lengths of the springs to vary between first and second fixed lengths, while in other embodiments, the length adjusting mechanisms may be configured to allow the effective lengths of the springs to be varied as desired, thus allowing for greater or lesser degrees of constraint of the movement of the wings **144** relative to the center section **142** of the header **104** during different non-harvesting transport and/or harvesting uses of the harvester **100**. Additionally, while in some such embodiments the length adjusting mechanisms of the wings **144** may be actuated by the suspension system **200** in tandem with one another, in other embodiments, the length adjusting mechanisms may be actuated independent of one another, such that the degree to which movement of the left wing **144a** is constrained may vary from the degree to which movement of the right wing **144b** is constrained, and vice versa.

(32) According to other embodiments, the suspension system **200** may alternatively, or additionally, comprise one or more elements (such as, e.g., variable length legs) that may be selectively engaged and/or actuated to physically constrain the degree to which the wings **144** may deflect upwards and/or downward to a predetermined range of motion and/or block the wings **144** from deflecting upwards and/or downwards past a predetermined distance during different operational modes of the harvester. In some such embodiments, such physically constraining/blocking elements may be used in conjunction with the features of to any of variable spring rate suspension system **200** embodiments described herein. Alternatively, in other such embodiments in which the suspensions system **200** includes such constraining/blocking element, the suspension system **200** may be defined by a single fixed spring rate during operation of the harvester **100** according to any of the various operational modes of the harvester **100**.

(33) In yet other embodiments, the suspension system **200** may comprise a hydraulic system configured to provide for first and second variable states corresponding to the harvesting and non-harvesting transport modes of the suspension system **200** which selectively allow for differing degrees of resistance to the pivoting of the wings **144** relative to the center section **142**. For example, in some embodiments (not shown), the suspension system **200** may comprise a pair a hydraulic circuits that are operably provided for each of the left wing **144a** and right wing **144b**, with a first circuit having a different volume and/or pressure of fluid than a second, distinct circuit

defining the pair of hydraulic circuits, such that the suspensions system **200** is defined by differing degrees of stiffness depending on which hydraulic circuit the suspension system **200** is fluidly engaged to.

(34) Referring to FIG. 4, a simplified schematic of a hydraulic based spring suspension system **200** comprising a blocker valve **300** which is configured to provide for first and second variable states corresponding to harvesting and non-harvesting transport modes of the suspensions system **200**, respectively, according to one embodiment is illustrated. The suspension system **200** illustrated in FIG. 4 corresponds to a spring suspension system **200** configured for use with one of the left wing **144a** or right wing **144b** of the header **104**, with the other of the left wing **144a** or right wing **144b** being provided with a substantially similar, albeit mirrored, suspension system **200** as shown in and described with reference to FIG. 4.

(35) Suspension system **200** generally comprises a fluid cylinder **202** that is fluidly connected to one or more accumulators **206** via an attenuation hose **208**. The accumulators **206** are configured to store a volume of pressurized fluid (such as, e.g., incompressible hydraulic fluid) that is supplied to the fluid cylinder **202** via the attenuation hose **208**. As fluid flows into or out from the fluid cylinder **202**, the fluid cylinder **202** is configured to extend or retract. As the fluid cylinder **202** is configured to suspend the wing **144**, the retraction and extension of the fluid cylinder **202** in response to changes in the amount of fluid within fluid cylinder **202** causes the wing **144** to move pivotally about the center section **142**, resulting in the upward or downward movement of the wing **144** relative to the center section **142**.

(36) A blocker valve **300** is fluidly disposed between the fluid cylinder **202** and the accumulators **206**. As will be described with more detail with reference to FIGS. 6A-9B below, the blocker valve **300** is configured to allow for selective flow of fluid between the fluid cylinder **202** and the accumulators **206**, allowing the fluid cylinder **202** to provide varying degrees of suspension of the wing **144**, which in turn allows the suspension system **200** to provide for first and second variable states in which the suspension system **200** is defined having differing stiffnesses that selectively allow for differing degrees of resistance to the pivoting of the wings **144** relative to the center section **142** during harvesting and non-harvesting transport of the harvester **100**.

(37) As shown in FIG. 6A, the accumulators **206** are additionally fluidly connected to a hydraulic block **210**, which serves as a source of fluid for the accumulators **206**. Fluid from the hydraulic block **210** is supplied to the accumulators **206** in response to the selective activation of a valve **212** to permit flow between the hydraulic block **210** and accumulators **206**. Once sufficient fluid has been allowed to fill the accumulators **206** to a desired pressure, the valve **212** may be activated to a closed configuration. As will be understood, according to some embodiments, a single hydraulic block **210** may be common to the suspension systems **200** of both the left wing **144a** and the right wing **144b**, while in other embodiments, the suspension systems **200** of each of the left wing **144a** and the right wing **144b** may comprise distinct, individual hydraulic blocks **210**.

(38) Referring to FIGS. 5A-5C, a blocker valve **300** according to various embodiments is illustrated. In general, the blocker valve **300** is selectively actuatable between a flow position, defined by a flow structure **302** and a restricted flow position defined by a flow-restriction structure **304**. As will be understood, blocker valve **300** may be biased to either the flow position or restricted-flow position, and may be selectively energized or otherwise activated between the flow and restricted flow positions according to any number of different arrangements, including mechanical and/or electromechanical arrangements.

(39) Additionally, while in some embodiments the activation of the blocker valve **300** between the flow position and the restricted-flow position may be controlled directly by the operator as desired, according to other embodiments, the activation of the blocker valve **300** may be controlled by a control system of the harvester **100**. For example, according to some embodiments, the harvester **100** may comprise a control system, which, in addition to controlling other aspects of the operation of the harvester **100**, may additionally be configured to control the activation of the blocker valve

300. According to some such embodiments, the control system may be configured to automatically activate the blocker valve **300** to the restricted-flow position upon the control system exiting out of an auto-header height mode of the control system and/or in response to the feederhouse **108** (and attached header **104**) being lifted up and elevated with respect to the ground. In yet other embodiments, the control system may be configured such that, when the harvester **100** is operated in a manual mode, the blocker valve **300** is automatically activated to a restricted-flow position upon the control system receiving a signal from ground detection sensors that the header **104** has been elevated off of the ground.

(40) When the blocker valve **300** is in the restricted-flow position, fluid present within the fluid cylinder **202** and attenuation hose **208** is prevented from flowing into the accumulators **206**. However, according to some embodiments, it may be advantageous to allow for a limited degree of fluid flow from the accumulators **206** into the attenuation hose **208** and fluid cylinder **202**.

Accordingly, as shown in FIG. 5A, according to some embodiments, the flow-restriction structure **304** of blocker valve **300** may comprise an orificed check valve structure **400**, which is configured to restrict flow from the attenuation hose **208** and fluid cylinder **202** into the accumulators **206**, but which allows for flow from the accumulators **206** into the attenuation hose **208**, even when the blocker valve **300** is in the restricted flow position.

(41) In other embodiments, it may be desired that there be no flow in either direction (i.e. no flow of fluid into or out of the accumulators **206**) when the blocker valve **300** is in the restricted flow position. According to some such embodiments, the flow-restriction structure **304** of blocker valve **300** may comprise a double-checked valve structure (such as, e.g., illustrated in FIG. 5B) or other structure configured to prevent flow in either direction through the blocker valve **300**.

(42) As shown in FIG. 5C, in some embodiments in which the blocker valve **300** comprises a flow-restriction structure **304** configured to prevent flow in either direction through the blocker valve **300** (such as, e.g., a double-checked valve flow-restriction structure **304**), the suspension system **200** may include an orificed check valve structure **400** arranged fluidly in parallel with the blocker valve **300**. By providing an alternate fluid path through which fluid from the accumulators **206** may flow into the attenuation hose **208**, the orificed check valve structure **400** may allow for restricted flow of fluid from the accumulators **206** into the attenuation hose **208** even when the blocker valve **300** is in the restricted-flow position.

(43) The ability of the suspension system **200** to provide for first and second variable states which selectively allow for differing degrees of pivoting of the wings **144** relative to the center section **142** will now be described with reference to FIGS. 6A-9B.

(44) Referring to FIGS. 6A and 6B, a simplified block diagram of the suspension system **200** and the header **104** configuration is shown during harvesting operation of the harvester **100** according to some embodiments. As described above, during harvesting, the articulated configuration of header **104** (in which left wing **144a** is hingedly attached to center section **142** via a left hinge joint **146a** and in which right wing **144b** is hingedly attached to center section **142** via a right hinge joint **146b**) allows the wings **144** of the header **104** to pivot upward and/or downward relative to the center section **142** to allow the header **104** to more closely follow the contours of the terrain.

(45) As shown in FIG. 6A, to facilitate the ability of the wings **144** to travel over and follow contours in terrain during harvesting operation, the blocker valve **300** is in a flow configuration during the harvesting mode of the suspension system **200**, in which the flow structure **302** of the blocker valve **300** is aligned between the accumulators **206** and the attenuation hose **208** so as to allow fluid to freely flow between the accumulators **206** and fluid cylinder **202**. As described above, by allowing fluid to flow into and out from the fluid cylinder **202** the fluid cylinder **202** is able to extend and retract as needed in response to changes in terrain. As will be understood, according to various embodiments, a float system configured to assist the header **104** in adapting to changes in terrain (such as, e.g., by monitoring changes in pressure imparted onto the header **104** and assisting in the flow of fluid into and out from the fluid cylinder **202** so as to maintain a desired

target pressure) may be incorporated into harvester **100**.

(46) Because fluid is allowed to flow freely between the accumulators **206** and fluid cylinder **202** during harvesting operation of the device, the pressure within the attenuation hose **208** will be substantially the same as the pressure within the accumulators **206**. The simplified block diagram of FIG. **6B** illustrates the header **104** when the header **104** is positioned on substantially flat terrain, and as such, the entire header **104** is shown in FIG. **6B** as extending in a generally planar manner. However, as will be understood, if the simplified block diagram of FIG. **6B** were to represent the header **104** along uneven terrain, the wings **144** of header **104** would be shown as extending substantially parallel to the terrain above which the wings **144** extended, such that the wings **144** would extend at non-zero degree angles relative to the center section **142**.

(47) Referring to FIG. **7A**, a simplified diagram of the suspension system **200** according to one embodiment is illustrated representative of a transition configuration of the harvester **100**, in which the header **104** is still supported by the ground (i.e. the header **104** has not been elevated to a point where the combine **102** supports the entirety of the weight of the header **104**) and in which the blocker valve **300** has been deenergized or otherwise deactivated from the flow position to the restricted-flow position.

(48) In the transition configuration, the switching of the blocker valve **300** into the restricted-flow position prevents any fluid from flowing into or out from the accumulators **206**. Upon entering into the transition configuration, the amount of fluid within the fluid cylinder **202** and attenuation hose **208** corresponds to the amount of fluid that had been present within the fluid cylinder **202** and attenuation hose **208** immediately prior to the blocker valve **300** being switched to the restricted-flow position. Accordingly, upon entering the transition configuration (i.e., upon initiation of the non-harvesting transport mode of the suspensions system **200**), the wings **144** are 'locked' in their last position prior to the harvester **100** being put into the transition configuration. The 'locked' configuration of the wings **144** may correspond to a configuration of the wings **144** in which one or both of the wings **144** extend angled upward relative to center section **142**, extend angled downward relative to center section **142**, and/or extend substantially parallel to center section **142**. As will be understood, the configuration of the wings **144** in the 'locked' position will depend on whether the fluid cylinder **202** was in a retracted, expanded, or neutral state immediately prior to switching the blocker valve **300** into the restricted-flow configuration.

(49) As illustrated in FIG. **7B**, because the header **104** remains partially supported by the ground in the transition configuration, the wings **144** of the header **104** remain extending in a direction substantially parallel to the terrain above which the wings **144** are supported. As similarly described with reference to FIG. **6B**, the header **104** that is represented by the simplified block diagram of FIG. **7B** is shown in a configuration in which the header **104** is positioned atop substantially horizontal terrain. However, as will be understood, if the simplified block diagram of FIG. **7B** were to represent the header **104** positioned along uneven terrain, the wings **144** of header **104** would be shown as extending substantially parallel to the surface above which the wings **144** extend, such that the wings **144** would extend at non-zero degree angles relative to the center section **142**.

(50) Referring to FIG. **8A**, a simplified block diagram representative of the suspension system **200** during non-harvesting transport of the harvester **100** with the header **104** in an elevated position in which the weight of the header **104** is entirely supported by the combine **102** is shown according to one embodiment. As shown in FIG. **8A**, in such an elevated, non-harvesting configuration of the header **104**, the blocker valve **300** remains closed in a restricted-flow configuration, in which flow of fluid from the fluid cylinder **202** and attenuation hose **208** into the accumulators **206** is prevented. According to various embodiments, in order to decrease the degree of stiffness of the suspension system **200** when in the non-harvesting mode, the attenuation hose **208** may be constructed with a desired degree of elasticity and resilience, which allows the attenuation hose **208** to expand to hold increased volumes of fluids as compared to an initial, neutral configuration of the

attenuation hose **208**. Although the flow of fluid into the accumulators **206** is prevented by the blocker valve **300**, fluid is free to flow between the fluid cylinder **202** and attenuation hose **208** during the elevated, non-harvesting transport configuration of the header **104**. As such, when the header **104** is elevated, causing the wing **144** to no longer be supported the ground, the elastic nature of the attenuation hose **208** is configured to allow some, or all, of the fluid that was 'locked' in the fluid cylinder **202** during the transition configuration (as described with reference to FIGS. 7A and 7B above) to flow into the attenuation hose **208**, thereby increasing the volume of 'locked' fluid already present within the attenuation hose **208** (as also described with reference to FIGS. 7A and 7B above).

(51) As representatively illustrated by the simplified block diagram of FIG. 8A, the displacement of some or all of the fluid from the fluid cylinder **202** into the attenuation hose **208** increases the pressure of the fluid within the attenuation hose **208** to a pressure that is greater than the pressure of the fluid stored within the accumulators **206**. Meanwhile, as representatively illustrated by the simplified block diagram of FIG. 8B, the decrease in the volume of fluid within the fluid cylinder **202** resulting from the displacement of fluid from the fluid cylinder **202** into the attenuation hose **208** decreases the ability of the fluid cylinder **202** to suspend the wing **144**, which in turn causes the wing **144** to pivot downward relative to the center section **142** by an angle of α_1 from an initial wing **144** position defined by the position of the wing **144** in the transition configuration (which in turn, corresponds to last position of the wing **144** during the last harvesting configuration of the header **104** prior to the blocker valve **300** being switched to a restricted-flow position).

(52) According to various embodiments, the angle α_1 may range from approximately 0.05° to 1.5° , more specifically between approximately 0.5° and 1.0° , and even more specifically between approximately 0.6° and 0.8° . As will be understood, the angle α_1 by which the left wing **144a** is pivoted downwards relative to the center section **142** during the elevated, non-harvesting transport configuration may be the same or may be different than the angle α_1 by which the right wing **144b** is pivoted downwards relative to the center section **142** during the elevated, non-harvesting transport configuration.

(53) Although, as shown in FIG. 8B, the wings **144** of the header **104** will exhibit some degree of sagging (i.e. pivoting of the wings **144** downwards relative to the center section **142**), with respect to the initial position of the wings **144** as defined by the position of the wings **144** during the transition configuration, the position of the wings **144** during the elevated, non-harvesting transport configuration may extend at an upwards angle relative to the center section **142**, generally planar with the center section **142**, or at a downwards angle relative to the center section **142**. As will be understood, the angle(s) relative to the center section **142** at which the wings **144** extend during the elevated, non-harvesting transport configuration will depend on factors including the angle(s) of the wings **144** relative to the center section **142** during the last harvesting configuration of the header **104** prior to the blocker valve **300** being switched to a restricted-flow position as well as the angle(s) α_1 by which the wings **144** are pivoted downwards during the elevated, non-harvesting transport configuration.

(54) According to various embodiments, as the harvester **100** is in the elevated, non-harvesting transport configuration (such as, e.g., represented in FIGS. 8A and 8B), the harvester **100** may transition to a downward flex configuration in response to the mass of the header **104** being subject to a downwards acceleration force (such as, e.g., in response to the harvester **100** travelling over uneven terrain). As illustrated in FIG. 9A, during such additional loading of the header **104** in the downward flex configuration, additional fluid flows out of the fluid cylinder **202** and into the attenuation hose **208**. This additional fluid causes the volume of fluid within the attenuation hose **208** to further increase from the increased volume that the attenuation hose **208** was subject to during the elevated, non-harvesting transport configuration. As shown in FIG. 9A, as a result of this additional fluid now held within the attenuation hose **208**, the pressure within the attenuation hose **208** is further increased.

(55) Meanwhile, as representatively illustrated by the simplified block diagram of FIG. 9B, the additional decrease in the volume of fluid within the fluid cylinder **202** as fluid is displaced from the fluid cylinder **202** and into the attenuation hose **208** during the downward flex configuration causes the wing **144** to pivot further downwards relative to the center section **142** by an angle of a_2 . According to various embodiments, the angle a_2 may range from approximately 0.05° to approximately 2.0° , more specifically between approximately 0.5° and approximately 1.5° , and even more specifically by approximately 1.0° . As will be understood, the angle a_2 by which the left wing **144a** is pivoted downwards relative to the center section **142** during the downward flex configuration may be the same or may be different than the angle a_2 by which the right wing **144b** is pivoted downwards relative to the center section **142** during the downward flex configuration.

(56) As explained with reference to FIG. 8B, although, as shown in FIG. 9B, the wings **144** of the header **104** will exhibit some degree of sagging (i.e. pivoting of the wings **144** downwards relative to the center section **142**), with respect to the position of the wings **144** in the configuration immediately prior to the downward flex configuration of the header **104** (such as, e.g., the elevated, non-harvesting transport configuration of FIGS. 8A and 8B), the position of the wings **144** during the downward flex configuration may extend at an upwards angle relative to the center section **142**, generally planar with the center section **142**, or downwards relative to the center section **142**. As will be understood, the angle(s) relative to the center section **142** at which the wings **144** extend during the downward flex configuration will depend on factors such as, e.g., the angle(s) of the wings **144** relative to the center section **142** during the last harvesting configuration of the header **104** prior to the blocker valve **300** being switched to a restricted-flow position; the angle(s) a_1 by which the wings **144** are pivoted downwards during the elevated, non-harvesting transport configuration; the angle(s) a_2 by which the wings **144** are pivoted downwards during the downward flex configuration; etc.

(57) As illustrated by FIGS. 6A-9B, the ability of the blocker valve **300** to isolate flow into the accumulators **206** during a restricted-flow position and to allow flow to and from the accumulators **206** during a flow position provides the suspension system **200** with first and second variable states defined by varied spring rates, which selectively provide differing degrees of resistance to the movement of the wings **144**, and thus allow for differing degrees of pivoting of the wings **144** relative to the center section **142**. As discussed with reference to FIGS. 6A and 6B, when the blocker valve **300** is in the flow position, the first variable state (i.e. the harvesting mode) is defined by the hydraulic circuit defined between the fluid cylinder **202**, the accumulators **206**, and the attenuation hose **208**. In this first variable state, the ability of fluid to flow freely between the fluid cylinder **202** and the accumulators **206** provides the hydraulic circuit with a relatively low stiffness, and thus allows the wings **144** to pivot about the center section **142** by an amount that defines a first range of motion. By allowing the wings **144** to pivot about the center section **142**, the suspension system **200** enables the wings **144** to dynamically adapt to and follow terrain, which, in addition to increasing crop yield efficiency, also reduces the dynamic loads on the harvester **100** during harvesting operation.

(58) As discussed with reference to FIGS. 7A-9B, when the blocker is in the restricted-flow position, the second variable state (i.e. the transport mode) is defined by the hydraulic circuit defined between the fluid cylinder **202** and the attenuation hose **208**. In the second variable state, the expandable nature of the attenuation hose **208** allows the attenuation hose **208** to hold fluid that may flow out of the fluid cylinder **202**. This ability of the attenuation hose **208** to hold an increased capacity of fluid provides the suspension system **200** with a manner by which the wings **144** are provided with a second range of motion by which the wings **144** may pivot relative to the center section **142**. However, overall, the hydraulic circuit in the transport mode is defined by a greater spring rate than when in the harvesting mode, and accordingly movement of the wings **144** relative to the center section **142** is subject to an increased resistance that results in the movement of the wings **144** being limited to a smaller range of motion than in the first state.

(59) Because the second range of motion is smaller than the first range of motion (such as, e.g., by between approximately 10% and approximately 30%), the ability of the wings **144** to pivot about the center section **142** is more limited when the suspension system **200** is in the second variable state than when the suspension system **200** is in the first variable state. As such, when the header **104** is elevated from the ground with the suspension system **200** in the second variable state (such as, e.g., discussed with reference to the elevated, non-harvesting transport configuration shown in FIGS. **8A** and **8B**) the suspension system **200** is configured to maintain the header **104** in a relatively level configuration in which the header **104** only exhibits a minimum amount of sagging, thus minimizing the risk of the outermost ends of the wings **144** inadvertently coming into contact with the ground during non-harvesting transport of the harvester **100**.

(60) Although the range of motion through which the wings **144** are able to pivot in the second variable state is limited, by providing even a limited range of motion by which the wings **144** are able to pivot about the center section **142**, (such as, e.g., by a range of between approximately $\pm 0.05^\circ$ and approximately $\pm 2.0^\circ$) the suspension system **200** is able to reduce the mass of the header **104** that is accelerated during transport of the harvester **100** (such as, e.g., while the harvester is in the elevated, non-harvesting transport configuration), thereby reducing the stress on the structure of the combine **102** (such as, e.g., by at least approximately 5%).

(61) As noted above, the ability of the suspension system **200** to provide the wings **144** with a limited ability to flex while the suspension system **200** is in the second variable state is provided by the ability of the attenuation hose **208** to hold fluid that flows out from the fluid cylinder **202** when the wings **144** are subject to downward forces (such as, e.g., when the header **104** is elevated entirely off of the ground in the elevated, non-harvesting transport configuration or during the downward flex configuration in which the harvester **100** travelling with an elevated header **104** encounters uneven terrain). Accordingly, as will be understood, in various embodiments, the range of motion through which the wings **144** are able to pivot while the suspension system **200** is in the second variable state may be varied by changing any number of variables that would affect the spring constant of the hydraulic circuit in the transport mode, such as, e.g., changing the length of the attenuation hose **208**, changing the selection of materials and/or structure of the attenuation hose **208** (to either make the attenuation hose **208** more or less compressible), etc. Additionally, according to some embodiments, the suspension system **200** may optionally be provided with an additional structure via which fluid may be added to and/or removed from the circuit defined by the fluid cylinder **202** and the attenuation hose **208** when the suspension system **200** is in the second variable state.

(62) As will be understood, although the harvesting configuration of FIGS. **6A** and **6B**, the transition configuration of FIGS. **7A** and **7B**, the elevated, non-harvesting transport configuration of FIGS. **8A** and **8B**, and the downward flex configuration of FIGS. **9A** and **9B** have been described as occurring in a sequential manner, as will be understood, the various configurations illustrated and described with reference to FIGS. **6A-9B** may occur according to any other number of sequences, in which any of the configurations may be repeated any number of different times. Additionally, the header **104** may be subject to the configurations of FIGS. **6A-9B** for varied durations of time. For example, according to various embodiments, the downward flex configuration of FIGS. **9A** and **9B** may directly follow the transition configuration of FIGS. **7A** and **7B**. In some embodiments, the transition configuration of FIGS. **7A** and **7B** may directly follow the elevated, non-harvesting transport configuration of FIGS. **8A** and **8B**.

(63) As will also be understood, although the articulated header **104** illustrated herein has been shown as comprising three sections: a center section **142**, a left wing **144a**, and a right wing **144b**, according to other embodiments, the articulated header **104** may comprise any number of different sections, including, e.g., a two section arrangement defined by only the left wing **144a** and the right wings **144b**.

(64) The construction and arrangement of the systems and methods as shown in the various

exemplary embodiments are illustrative only. Although only a few embodiments have been described in detail in this disclosure, many modifications are possible (e.g., variations in sizes, dimensions, structures, shapes and proportions of the various elements, values of parameters, mounting arrangements, use of materials, colors, orientations, etc.). For example, the position of elements may be reversed or otherwise varied and the nature or number of discrete elements or positions may be altered or varied. Accordingly, all such modifications are intended to be included within the scope of the present disclosure. References herein to the positions of elements (e.g., “top,” “bottom,” “above,” “below,” “upper,” “lower”, etc.) are merely used to describe the orientation of various elements in the FIGURES. It should be noted that the orientation of various elements may differ according to other exemplary embodiments, and that such variations are intended to be encompassed by the present disclosure.

(65) The term “coupled,” as used herein, means the joining of two members directly or indirectly to one another. Such joining may be stationary (e.g., permanent or fixed) or moveable (e.g., removable or releasable). Such joining may be achieved with the two members coupled directly to each other, with the two members coupled to each other using a separate intervening member and any additional intermediate members coupled with one another, or with the two members coupled to each other using an intervening member that is integrally formed as a single unitary body with one of the two members. Such members may be coupled mechanically, electrically, and/or fluidly. The order or sequence of any process or method steps may be varied or re-sequenced according to alternative embodiments. Other substitutions, modifications, changes, and omissions may be made in the design, operating conditions and arrangement of the exemplary embodiments without departing from the scope of the present disclosure.

Claims

1. A harvester system comprising: a header comprising a first section; and a suspension system comprising: a first fluid cylinder coupled to the first section; a first attenuation hose; a first accumulator; and a first valve selectively openable to permit flow between the first attenuation hose and the first accumulator; wherein, when the suspension system is operated in a first mode, pivoting movement of the first section is subject to a first degree of resistance provided by the suspension system, and wherein, when the suspension system is operated in a second mode, pivoting movement of the first section is subject to a second degree of resistance provided by the suspension system.
2. The harvester system of claim 1, wherein the first valve is open during operation of the suspension system in the first mode and is closed during operation of the suspension system in the second mode, and wherein the suspension system is defined as having a first stiffness in the first mode and a second stiffness, greater than the first stiffness, in the second mode.
3. The harvester system of claim 2, wherein a first channel extends through the first attenuation hose, and the first channel defines a first volume in the first mode and defines a second volume greater than the first volume during at least a portion of operation of the suspension system in the second mode.
4. The harvester system of claim 3, wherein a cross-sectional area of the first channel is greater during the second mode than a cross-sectional area of the first channel during the first mode.
5. The harvester system of claim 2, further comprising a fluid block, wherein the first accumulator is fluidly connected to the fluid block during at least the first mode.
6. The harvester system of claim 1, wherein the header further comprises a second section and a center section to which each of the first section and the second section is attached, each of the first section and second section being independently pivotable relative to the center section.
7. The harvester system of claim 6, wherein the suspension system includes: a second fluid cylinder coupled to the second section; a second attenuation hose; a second accumulator; and a second valve

selectively openable to permit flow between the second attenuation hose and the second accumulator, and wherein one of or both of the first attenuation hose or the second attenuation hose includes a predetermined degree of elasticity configured to enable the first attenuation hose or the second attenuation hose to expand and to hold an increased volume of fluid compared to an initial, neutral configuration of the first attenuation hose or the second attenuation hose.

8. The harvester system of claim 7, wherein the first valve includes a first blocker valve and the second valve includes a second blocker valve, and wherein each of the first blocker valve and the second blocker valve includes a flow position and a restricted flow position.

9. The harvester system of claim 8, wherein the flow position selectively provides a first range of motion to the first section and to the second section relative to the center section located between the first section and the second section.

10. The harvester system of claim 9, wherein the restricted flow position provides a second range of motion to the first section and to the second section relative to the center section.

11. The harvester system of claim 10, wherein the first range of motion is configured to enable a flow of fluid freely between the first fluid cylinder and the first accumulator, and freely between the second fluid cylinder and the second accumulator to provide the suspension system with a first stiffness.

12. The harvester system of claim 11, wherein the second range of motion is configured to restrict a flow of fluid between the first fluid cylinder and the first accumulator and to restrict a flow of fluid between the second fluid cylinder and the second accumulator to provide the suspension system with a second stiffness greater than the first stiffness, and wherein the movement of the first section and the second section in the second range of motion is limited to a smaller range of motion than the first range of motion.

13. The harvester system of claim 12, wherein the predetermined degree of elasticity enables each of the first attenuation hose and the second attenuation hose to provide the second range of motion with an increased resistance in the movement of the first section and the second section when compared to the first range of motion.

14. A harvester system comprising: a header comprising a first section; and a suspension system comprising: a first fluid cylinder coupled to the first section; a first attenuation hose; a first accumulator; and a first valve selectively openable to permit flow between the first attenuation hose and the first accumulator, wherein the first section is pivotable in each of a first mode and a second mode, wherein the first accumulator is in fluid communication with the first fluid cylinder, and wherein the first attenuation hose fluidly couples the first fluid cylinder and the first accumulator.

15. The harvester system of claim 14, wherein the first valve includes a first blocker valve and wherein the first blocker valve includes a flow position and a restricted flow position.

16. The harvester system of claim 15, wherein the flow position selectively provides a first range of motion to the first section relative to a center section located adjacent to the first section.

17. The harvester system of claim 16, wherein the restricted flow position provides a second range of motion of the first section relative to the center section.

18. The harvester system of claim 17, wherein the first attenuation hose includes a predetermined degree of elasticity configured to enable the first attenuation hose to expand and to hold an increased volume of fluid compared to an initial, neutral configuration of the first attenuation hose.

19. The harvester system of claim 18, wherein the first range of motion is configured to enable a flow of fluid freely between the first fluid cylinder and the first accumulator to provide the suspension system with a first stiffness.

20. The harvester system of claim 19, wherein the predetermined degree of elasticity enables the first attenuation hose to provide the second range of motion with a second stiffness, greater than the first stiffness, in the movement of the first section when compared to the first range of motion.
